

```

+-----+-----+
|                                     |
|          BUS INTERFACE UNIT (BIU)  |
|                                     |
| Segment Registers: [ CS ][ DS ][ SS ][ ES ]   IP: [ Instruction Ptr ] |
|                                     |
|                                     |
|          v                          |
|          20-Bit Adder (Physical Addr)         |
|                                     |
|                                     |
|          v                          |
|          6-Byte Instruction Stream Queue [Q1|Q2|Q3|Q4|Q5|Q6] |
|-----+-----+
|                                     |
|          Internal Bus               |
|-----+-----+
|                                     |
|          EXECUTION UNIT (EU)       |
|                                     |
| General Registers: [ AH | AL ] [ BH | BL ] [ CH | CL ] [ DH | DL ] |
| Pointers & Indexes: [ SP ] [ BP ] [ SI ] [ DI ] |
|                                     |
| Control Logic <---> 16-Bit ALU <---> Status & Control Flags (16-bit) |
|-----+-----+

```

2. Ex
and up

* AX

* AD

Q4. Explain memory segmentation and physical address calculation in 8086 with an example.

Physical Address Formula

Numerical Example

* Result: The 20-bit address generated on the physical address bus is 25124H.

Q5. Explain the functioning of 8086 in minimum mode with a diagram.

```

+-----+ ALE +-----+ -> A0 - A19 Address Bus
| 8086 CPU | ----> | Latches (74LS373)|
| (MN/MX' = +5V) | +-----+
|          | AD0-AD15 +-----+ -> D0 - D15 Data Bus
|          | -----> | Transceivers |
|          |          +-----+
|          | Control +-----+
|          | -----> | Memory / I/O |
+-----+ (RD,WR) +-----+
    
```

* In Minimum Mode (MN/MX' = 1), the 8086 operates as a standalone processor.

Q6. Explain the functioning of 8086 in maximum mode with a diagram.

```

+-----+ S0', S1', S2' +-----+ -> System Control Signals
| 8086 CPU | -----> | 8288 BUS | (MRDC, MWTC, AMWC,
| (MN/MX' = GND) | | CONTROLLER | IORC, IOWC, INTA)
+-----+ +-----+
    
```

* In Maximum Mode (MN/MX' = 0), status outputs S0', S1', S2' are sent to an external Intel 8288 Bus Controller.

Q7. Draw and explain the timing diagram for a read operation in minimum mode.

```

Clock Cycles :   T1   |   T2   |   T3   |   T4   |
CLK           :  --__--__--__--__--__--__--__--
A16-A19       :  == Address == Status ==
AD0-AD15       :  == Address == High-Z == Data ==
ALE           :  --__--__--__--__--__--__--
RD            :  -----
DT/R          :  -----
    
```

* T1: ALE goes HIGH; 20-bit address output on AD0-AD15 and A16-A19.

Q8. Draw and explain the timing diagram for a write operation in minimum mode.

```

Clock Cycles :   T1   |   T2   |   T3   |   T4   |
CLK           :  --__--__--__--__--__--__--
A16-A19       :  == Address == Status ==
AD0-AD15       :  == Address == Write Data ==
ALE           :  --__--__--__--__--__--
    
```

WR	:	-----
DT/R	:	-----

* T1: ALE goes HIGH; target address output on AD0-AD15.

* T2

Q9. Explain interrupt structure and interrupt servicing in 8086.

1. Interrupt Types:

* Ha